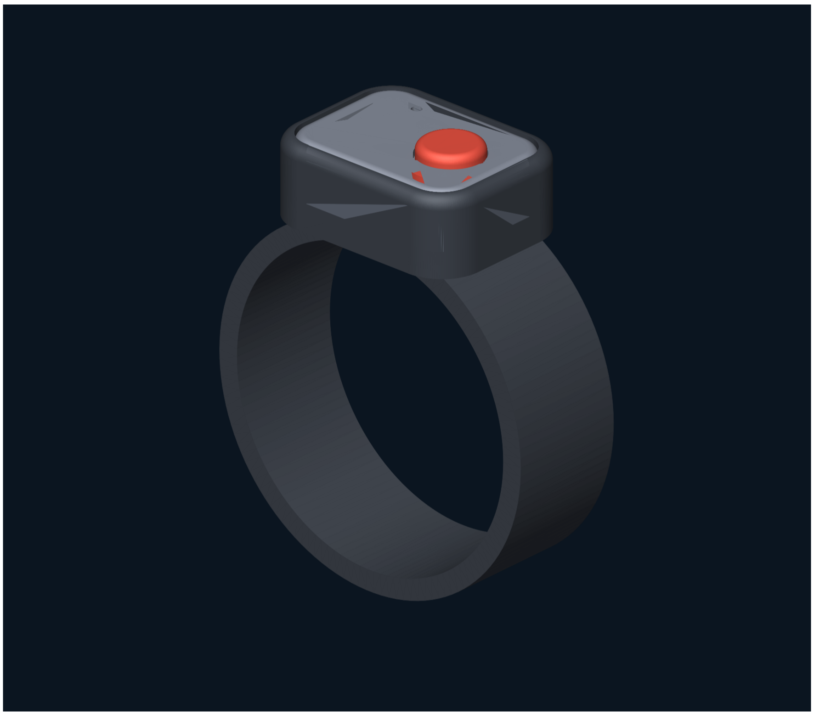
Dive Buddy Alert Band

Underwater buddy communication.



Paired calls. Tactile distance feedback.

A proposed slip-on silicone wristband calls a paired diver underwater, then turns a measured gap into a changing vibration cadence. One wrist transducer provides **distance, not direction**.

Product concept

A separate wristband uses one top-mounted button and a haptic motor for paired calls and distance feedback. Recreational buddy pairs are the intended users; instructors and dive shops are potential sales channels. Customer demand remains unvalidated.

An aid to buddy procedures, never a safety guarantee. The concept cannot identify a safe route, confirm visual contact or ensure reunion.

User interaction

Proposed pairing, call, ranging and stop sequence.

1 / Pair on the surface

Hold both buttons for 3 seconds; synchronized LEDs are the proposed pairing indication. A shared 16-bit code is an assumption, not proof of cross-pair isolation.

2 / A calls B

A short press requests three strong buzzes and a blinking LED on B. Packet acknowledgement and human acknowledgement are different; delivery feedback and retries remain undefined.

3 / A requests ranging

A 1-second hold starts the proposed exchange. Both bands participate in messaging; A receives the changing cadence. B's ongoing indication is still an open interaction decision.

4 / Stop

One press on either band requests Stop. Rev A proposes a five-minute timeout and a one-second burst then stop below 2 m. Neither proximity nor an automatic stop confirms reunion.

5 / Charge

Place the bands on an inductive dock. No band charging port. Dock power input, alignment, thermal design and completion indication need definition.

Use case

A buddy facing away receives a paired tactile call. During ranging, the caller receives a cadence that changes with the estimated separation. The proposed product combines addressed attention and tactile distance feedback in a standalone wristband.

Distance feedback

Distance feedback is not a directional navigation display.

Over 15 m / valid reply - One pulse every 4 seconds.

10-15 m - One pulse every 2 seconds.

5-10 m - One pulse every second.

2-5 m - A double pulse every 0.5 seconds.

Below 2 m - One-second burst, then stop. Near does not mean found.

No reply - Distance unknown. The revised demo suppresses proximity pulses; the device's distinct lost-link indication is an open requirement.

The acoustic exchange

Approximately 65 kHz ultrasonic two-way time-of-flight. Proposed distance calculation: **d = (round-trip time - fixed reply delay) x sound speed / 2**. Using 1,500 m/s and an assumed 2 ms reply delay, 20 m corresponds to about 26.7 ms propagation round trip and 28.7 ms total exchange time.

20 m range, approximately ±0.3 m distance performance and nominal 1 Hz exchange are proposed targets or assumptions, not demonstrated results. Timing resolution alone does not establish ranging accuracy. Body shadowing, multipath, orientation and environmental variation remain unresolved.

Demonstrator behavior

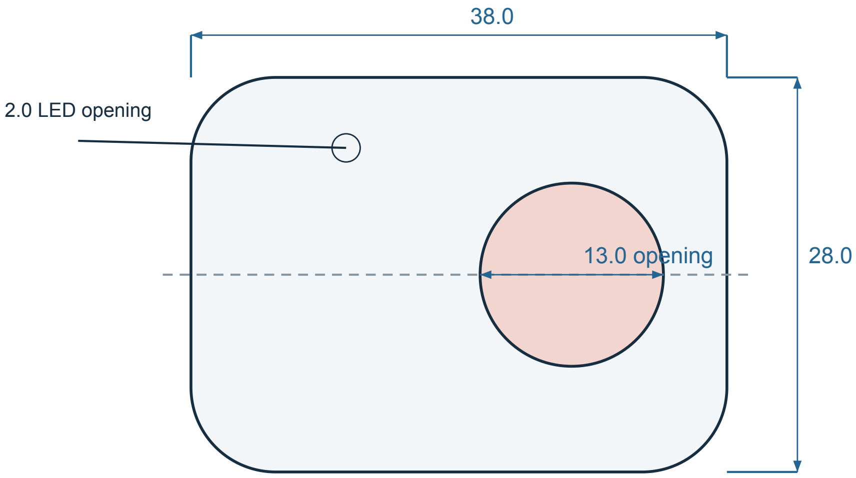
The web page provides Call, Start ranging, Stop, a simulated-distance control and a reply toggle. Its 20 m cutoff is an illustrative envelope, not a propagation model. Optional phone vibration illustrates cadence only. Production boundary hysteresis, actuator waveforms and user perceptibility are unspecified.

Dimensions & form

Nominal values from the existing source. No manufacturing tolerances are released.

MODULE / TOP VIEW

Housing envelope excludes button protrusion.

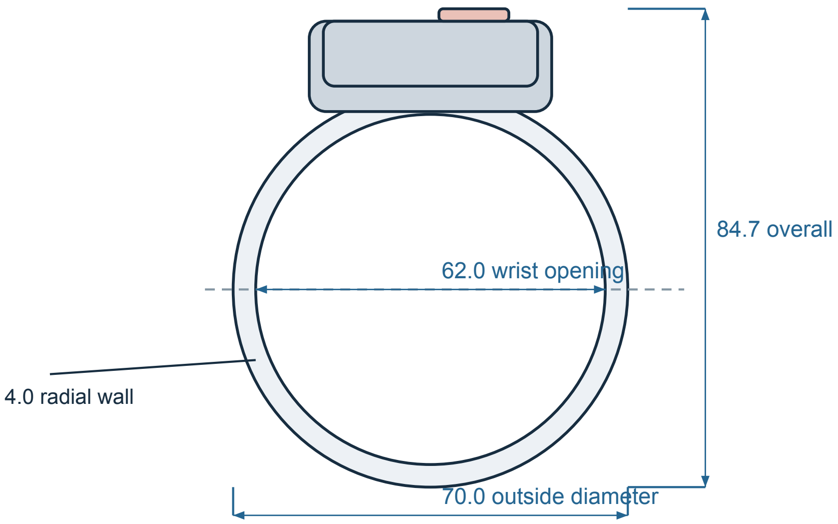


R6 nominal corners; button stem modeled at 12.4 diameter.

DBA / REV B | NOMINAL mm | CONCEPT ONLY - NOT TO SCALE

BAND / FRONT VIEW

Existing source geometry, with the module assembled.
24.0 band width along wrist axis; 33.0 cradle width.



DBA / REV B | NOMINAL mm | CONCEPT ONLY - NOT TO SCALE

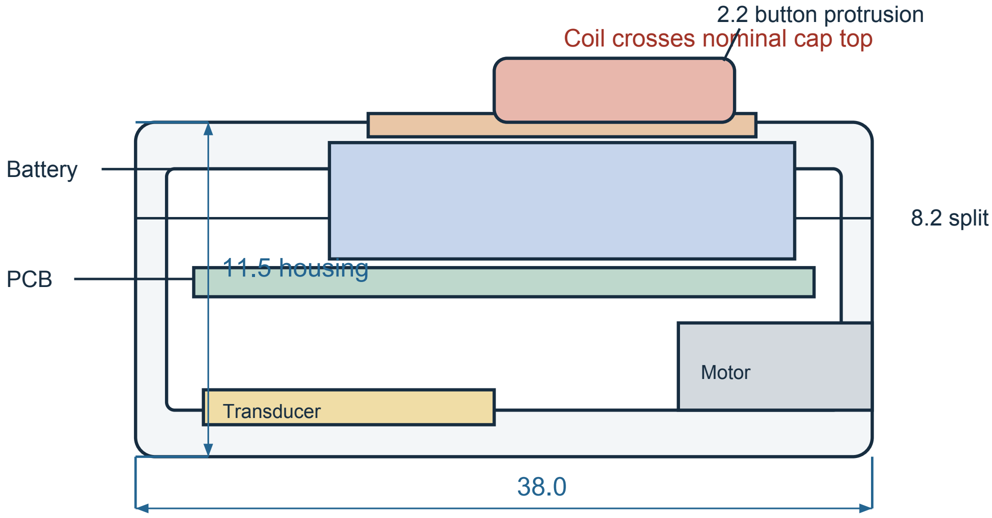
The 11.5 mm housing excludes the 2.2 mm modeled button protrusion. The source gives 84.7 mm overall assembly height including the button; the original DBA-001 sheet shows 88.5 mm. The dimension discrepancy remains an engineering review item.

Internal layout

Module section and component clearances.

MODULE / SCHEMATIC SECTION

Internal packaging is unresolved. This view does not certify fit.



Cap/battery overlap, coil clearance and switch/button stack require reconciliation.

DBA / REV B | NOMINAL mm | CONCEPT ONLY - NOT TO SCALE

Packaging interfaces

The source places the charging coil at 11.0-11.8 mm above the housing base; the nominal housing ends at 11.5 mm. The battery occupies 6.8-10.8 mm, extending into the nominal cap region. The switch/button stack also needs reconciliation. Wire routing, PCB keep-outs, ferrite, adhesives and battery allowance are not resolved.

Housing & charging

The module remains removable from the silicone band; that does not make the sealed cap user-openable. The button stays on top. Inductive charging through the cap avoids a band charging connector, with a dock, alignment and thermal design as trade-offs. The roughly two-hour charging time is a target. Sealing performance is not established by the charging choice.

Engineering requirements & open questions

Register 1-5 of 21. Design constraints, targets and pending engineering decisions.

Product scope | Fixed decision

One-to-one buddy attention and distance-feedback concept; an aid to buddy procedures, never a safety guarantee.

Unresolved: No promise of reunion, emergency rescue, bearing or navigation.
Discipline: Systems / product

Form and service | Fixed decision

Slip-on silicone band; sealed module removable from the band; button on top.

Unresolved: Retention force, release method, snag loads and actual wrist/cuff fit remain open. Removal from the band does not mean the sealed enclosure is user-openable.
Discipline: Mechanical / industrial design

Charging interface | Fixed decision

Inductive charging through the sealed cap; no charging port on the band.

Unresolved: Dock alignment, spacing, thermal limits, wet/salt-contaminated charging behavior and charge indication are unspecified. The dock may have its own power connector.
Discipline: Electronics / mechanical

Acoustic method | Fixed decision

Approximately 65 kHz ultrasonic two-way time-of-flight; one wrist transducer.

Unresolved: Transducer construction/resonance, directivity, drive level, link budget and receiver sensitivity need engineering definition. Component names in Rev A are candidates, not selections.
Discipline: Underwater acoustics

Distance and timing | Proposed target

20 m range; approximately ±0.3 m distance target; nominal 1 Hz exchange with assumed fixed 2 ms reply delay.

Unresolved: No measured performance supplied. Accuracy, resolution, latency and availability need separate definitions. Use d = (RTT - reply delay) x sound speed / 2; timing precision alone does not establish accuracy.
Discipline: Acoustics / firmware

Engineering requirements & open questions

Register 6-10 of 21. Design constraints, targets and pending engineering decisions.

Operating envelope | Proposed target

Rev A proposes 50 m operating depth, four 60-minute dives per charge and about 28 g assembled mass.

Unresolved: No depth rating or certification established. Define immersion duration, temperature, salinity, pressure convention, storage conditions and intended environments. The Rev A 6 bar note is ambiguous without gauge/absolute pressure.
Discipline: Systems / mechanical

Pairing | Rev A assumption

Hold both buttons 3 s on the surface; synchronized LEDs; a shared 16-bit pair code.

Unresolved: Address allocation, collisions, re-pairing, nearby-pair capacity and prevention of cross-pair alerts remain undefined. A code alone does not guarantee isolation.
Discipline: Firmware

Call and acknowledgement | Rev A assumption

Short press calls the buddy; receiver gives three strong buzzes and a blinking LED.

Unresolved: Distinguish packet acknowledgement from human acknowledgement. Retries, delivery feedback, busy states and whether the receiver must confirm are open.
Discipline: Firmware / interaction design

Ranging and stop | Rev A assumption

Caller holds 1 s to request ranging. Caller receives cadence; both bands exchange messages. One press on either band ends the session; nominal 5-minute timeout.

Unresolved: Receiver's ongoing indication, simultaneous calls, remote-stop confirmation and behavior after missed messages need a state specification.
Discipline: Firmware / interaction design

Haptic vocabulary | Rev A assumption

Greater than 15 m: pulse / 4 s; 10-15 m: pulse / 2 s; 5-10 m: pulse / 1 s; 2-5 m: double pulse / 0.5 s; below 2 m: 1 s burst, then stop.

Unresolved: Cadence is illustrative. Exact boundaries, hysteresis, glove/cuff perceptibility and actuator waveform remain open. Near is not visual contact or confirmed reunion.
Discipline: Interaction design / firmware

Engineering requirements & open questions

Register 11-15 of 21. Design constraints, targets and pending engineering decisions.

Lost link | Open question

A missing reply means distance is unknown; it must not be displayed as a valid far-away reading.

Unresolved: Rev A merges greater-than-15 m and no reply. Decide stale-data timeout, distinct indication and reacquisition behavior. The revised demo suppresses distance and proximity pulses when replies are disabled.
Discipline: Systems / firmware

Orientation and echoes | Open question

Body shadowing, wrist orientation, obstructions, other pairs and multipath may affect reception or range.

Unresolved: Define accepted environments, range-quality reporting, outlier handling and stale-reading policy. No directional arrow is provided to the diver.
Discipline: Underwater acoustics

Power budget | Rev A assumption

150 mAh, 3.7 V battery envelope; roughly 0.4 mA listening and 12 mA active-ranging estimates; roughly 2 h charge target.

Unresolved: Rev A's claimed approximately 100 mAh use does not reconcile with its state currents: 3 h 40 min x 0.4 mA + 20 min x 12 mA is about 5.5 mAh before other loads/losses. A complete duty-cycle budget is unresolved; runtime is not established.
Discipline: Electronics

Battery and charge behavior | Open question

Low-battery threshold, reserve behavior, cell protection, cold-water effects, aging allowance and charge completion are undefined.

Unresolved: Specify what the diver feels when power is insufficient for ranging or alerts. Capacity and package dimensions do not establish cell suitability.
Discipline: Electronics / firmware

Housing and seal | Rev A assumption

38 x 28 x 11.5 mm nominal housing, 1.6 mm nominal wall, 0.6 mm acoustic membrane; bonded or welded cap with secondary O-ring concept.

Unresolved: Material choice, joint, sealing details, pressure deflection, corrosion compatibility, leak paths, light-pipe/button sealing and acoustic coupling remain unresolved. No manufacturing tolerances are released.
Discipline: Mechanical / acoustics

Engineering requirements & open questions

Register 16-21 of 21. Design constraints, targets and pending engineering decisions.

Internal packaging | Open question

Existing source places the coil above the nominal cap envelope and the battery into the nominal cap region; the switch/button stack also needs review.

Unresolved: This is a layout illustration, not a cleared assembly. Resolve interference, wire routing, PCB keep-outs, ferrite, adhesive and battery allowance before treating it as an enclosure design.
Discipline: Mechanical / electronics

Band and button | Rev A assumption

62 mm relaxed wrist opening; 24 mm band width; 4 mm radial wall; 13 mm nominal button aperture; 3-4 N actuation target.

Unresolved: 95 mm stretched opening, one-size fit and glove operation are assumptions. BTN_H = 2.2 mm is modeled protrusion, not demonstrated switch travel. The visible stem is 12.4 mm in the source; flange is 16 mm.
Discipline: Industrial design / mechanical

Dimensions and drawing authority | Open question

Source gives 84.7 mm nominal assembled height including button; original DBA-001 shows 88.5 mm. Use the Rev B annotated concept views for presentation.

Unresolved: Rev A's blanket ±0.2 tolerance and drawing scale are not a release specification. SPLIT_Z = 3.2 is used with +5.0 in the source, yielding an 8.2 mm base. Do not infer parameter behavior from names alone.
Discipline: Mechanical

Commercial case | Hypothesis

Proposed first audience: recreational buddy pairs, reached through instructors and dive shops; paired sale with shared dock.

Unresolved: No customer interviews, purchase commitments, sales, market sizing or defensible novelty evidence supplied. The value of distance cadence over an attention-only alert remains a hypothesis.
Discipline: Product / commercial

Economics | Desk estimate

Rev A: approximately $43 per band / $86 per pair at 5,000 pairs; ±30% uncertainty; $179-199 per-pair retail target.

Unresolved: Not supplier quotes or demonstrated margins. Estimate excludes tooling, NRE, freight and duty. Channel terms, returns, support, warranty and complete delivered cost remain open.
Discipline: Commercial / manufacturing

Project status | Fixed decision

Concept by Rafael. Parametric layout and presentation assets are available. Hardware performance and manufacturing design are not established.

Unresolved: Feasibility, requirements and packaging interfaces require engineering review.
Discipline: Project lead

Pricing & commercial assumptions

Proposed paired kit, cost assumptions and development status.

Proposed offer

A paired kit and shared charging dock, with a Rev A retail target of **$179-199 per pair**. Recreational buddy pairs are the intended users; instructors and dive shops are candidate channels. No customer interviews, orders, purchase commitments or market sizing are supplied.

Cost basis

Rev A estimates approximately **$43 per band / $86 per pair** at a volume of 5,000 pairs, with **±30% uncertainty**. This is a desk estimate, not a supplier quote. The dock is allocated across the pair. Tooling, non-recurring engineering, freight and duty are excluded. Returns, warranty, support and channel terms remain open; retail minus this estimate is not established profit.

Product positioning

The intended combination is a simple slip-on form, a paired tactile call and distance-coded cadence, without a screen or exposed charging connector on the band. Its usefulness over simpler attention tools, and whether divers want another wrist device, remain hypotheses.

Development status

Concept definition, proposed user interaction and an eleven-part parametric layout are available. Acoustic performance, customer demand, manufacturability, certification and intellectual-property clearance remain unestablished.

Engineering scope

The specification covers underwater acoustics, electronic control and power, firmware states, sealing, module retention and user interaction. Component envelopes and nominal dimensions require engineering development before manufacturing release.

Design files

Concept specification, drawings and parametric source.

DBA_Interactive.html
Self-contained presentation content and original embedded mesh. Includes the narrative, interactive demo, projected dimensions, part callouts, drawing views and the requirements register. Three.js is loaded from its existing CDN.

DBA_Design_Package_RevB.pdf
Product concept, user interaction, technical specification and engineering review notes.

DBA_Annotated_Views_RevB.pdf
Three annotated concept views. Dimensions are nominal, not a manufacturing release.

cad/ and renders/
Unchanged original STEP, STL, DXF and render assets. The original drawing sheet is retained as a legacy reference; its discrepancies are recorded here.

dba_cad.py, render.py, drawing_sheet.py
Original source scripts, unchanged. Their original output path points to /home/claude/out; adapt the output location before any future authorized regeneration. No geometry has been regenerated in this revision.

presentation/
Editable presentation source, requirements register and generated annotation SVGs. The original int16 mesh payload is retained. Rebuild the presentation from these files.

legacy/DBA_Design_Package_RevA.pdf
Original reference preserved for traceability. Its ratings, performance wording, commercial claims and blanket tolerances are not established by the package.

Future geometry edits must begin in dba_cad.py, then regenerate CAD, renders, the drawing sheet and the HTML mesh. This revision changes presentation and annotations only.